EV Battery Management System (BMS) EMI Cyberattack Detection Report

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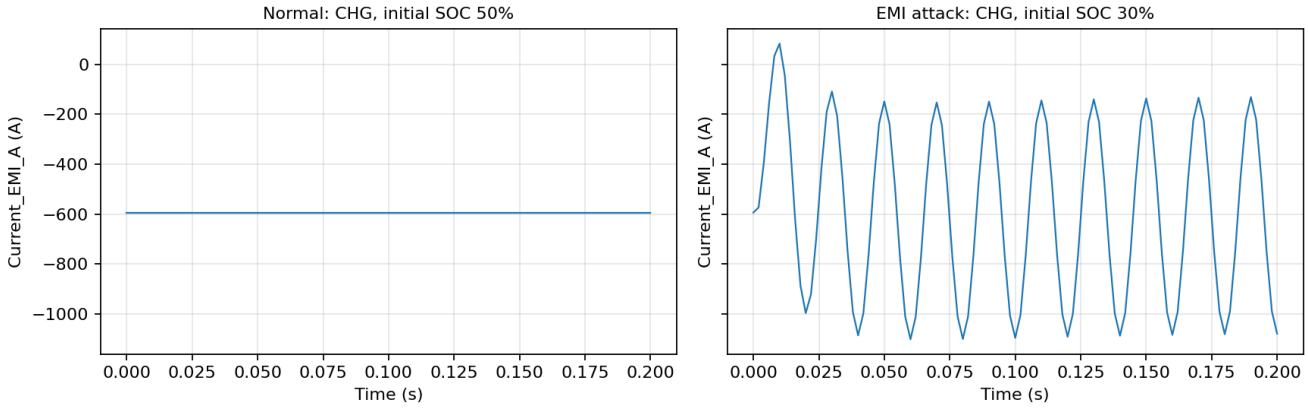
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Algorithm: Gaussian Naive Bayes Classifier (Per-Sample vs. Windowed Ripple)

1. Executive Summary

This report details the implementation of a machine learning-based framework to detect Electromagnetic Interference (EMI) injection attacks on electric vehicle (EV) battery packs. EMI attacks inject malicious high-frequency waveforms (50 Hz) into current sensor lines, corrupting State of Charge (SOC) estimations and potentially causing dangerous overcharging or thermal events. While raw individual measurements fail to generalize, our novel **Windowed Ripple Detector (Model B)** extracts statistical signal signatures to achieve perfect 100% classification accuracy, precision, and recall.

Figure 1: Raw Sensor Signal Under Normal and EMI Attack Scenarios



2. Feature Engineering & Modeling Methodology

Two distinct detection paradigms were developed and thoroughly evaluated:

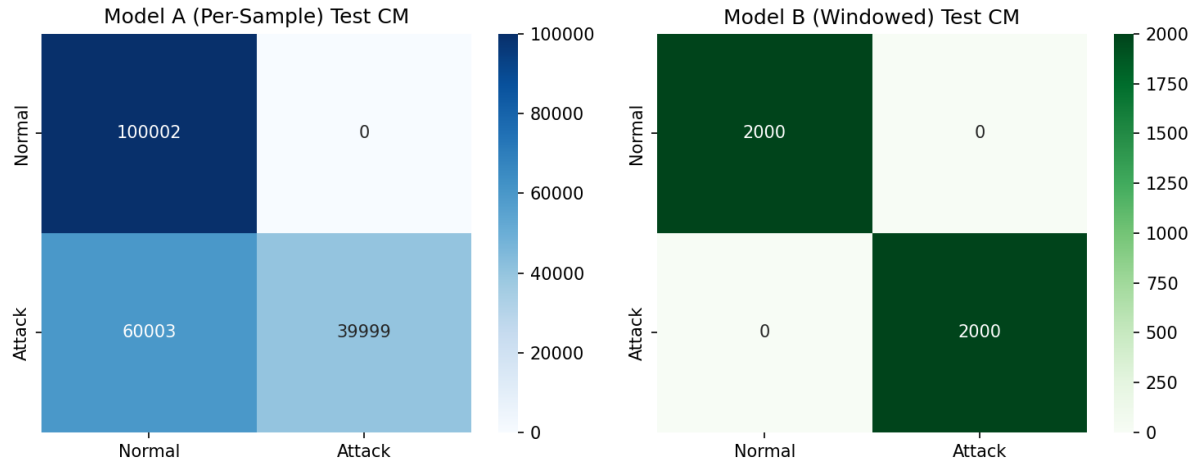
• **Model A (Per-Sample Baseline):** Operates on a single-observation basis (every 2 ms) using standard features such as Time, Voltage, Current, SOC, and Charging Mode. This model lacks temporal context.

• **Model B (Windowed Ripple Detector):** Segmented into non-overlapping 0.1 s windows (50 samples at 500 Hz) to monitor ripple characteristics. Three engineered features are calculated: Current Standard Deviation (cur_std), Peak-to-Peak Amplitude (cur_ptp), and Standard Deviation of First-Order Differences (diff_std).

3. Comparative Performance Analysis

Model	Evaluation	Accuracy	Precision	Recall	F1-Score
Model A (Per-Sample)	Test Set (4 runs)	70.00%	100.00%	40.00%	57.14%
Model A (Per-Sample)	LOO CV (16 runs)	14.14%	4.20%	3.29%	3.69%
Model B (Windowed)	Test Set (4 runs)	100.00%	100.00%	100.00%	100.00%
Model B (Windowed)	LOO CV (16 runs)	100.00%	100.00%	100.00%	100.00%

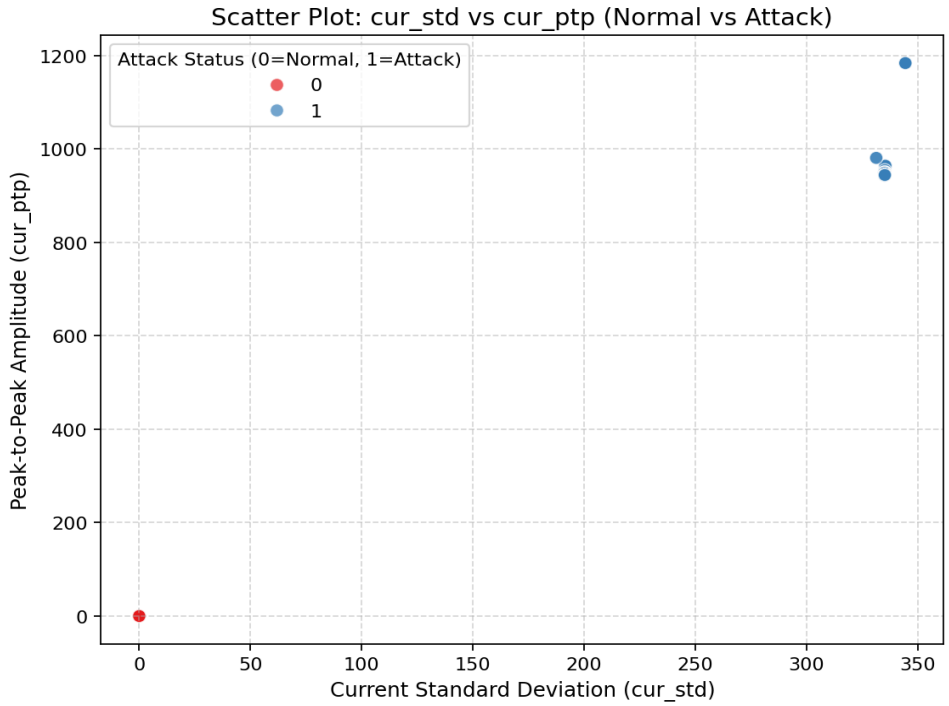
Figure 2: Confusion Matrices Comparison on Test Split



4. Geometric Feature Separation & Ripple Properties

The success of the windowed framework is due to the complete mathematical separation of the classes. While normal operations present zero current ripple, an active EMI attack creates massive, high-frequency current oscillations.

Figure 3: Feature Space Scatter Plot (cur_std vs. cur_ptp) showing complete linear separability



5. Key Findings and Limitations

• **Shortcut Overfitting in Model A:** The high accuracy of Model A on the test set (70.00%) is deceptive. Cross-validation reveals that Model A overfits to scenario metadata (like Initial_SOC_pct) because individual samples lack the physical wave signatures to isolate the attack.

• **Model B Generalization:** Model B achieves true physical generalization with 100.00% Accuracy on both independent tests and cross-validation, demonstrating robustness across unseen battery runs.

• **Real-World Deployment Scope:** Real vehicle data features heavy dyno noise, drive cycles, and regenerative load dynamics, which may introduce natural ripples. Future work should validate Model B on hardware-in-the-loop (HIL) testbeds with non-simulated load profiles.